

AIRWAY MALFORMATIONS AND FOREIGN BODIES

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Airway Malformations

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LARYNGOMALACIA :

- m/c congenital malformation of larynx.
- malacia : Softness/floppiness/laxity.
- Structures affected : **Supraglottic structures**
 - Epiglottis
 - Arytenoids
 - Aryepiglottic folds

Clinical manifestations :

Stridor : Sign of upper airway obstruction.

- Pathology : Floppy structures → Fall over glottic opening.
- Onset : Soon after birth/within 2 weeks after birth.
- **Characteristics**
 - Intermittent (When baby is crying/agitated/after feeding)
 - Postural variation : **Supine +** ; **Prone -**

Laryngomalacia vs Congenital Subglottic Stenosis :

	Laryngomalacia	Congenital subglottic stenosis
Positional variation of stridor	+	-
Cry	muffled	Normal
Stridor	Inspiratory	Biphasic (Inspiration and Expiration)

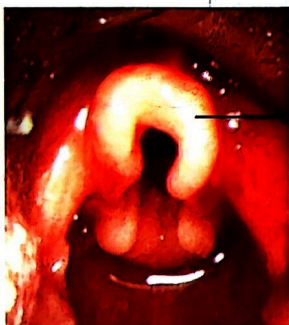
khanirfan0392@gmail.com

management :

Diagnosis : Laryngoscopy.

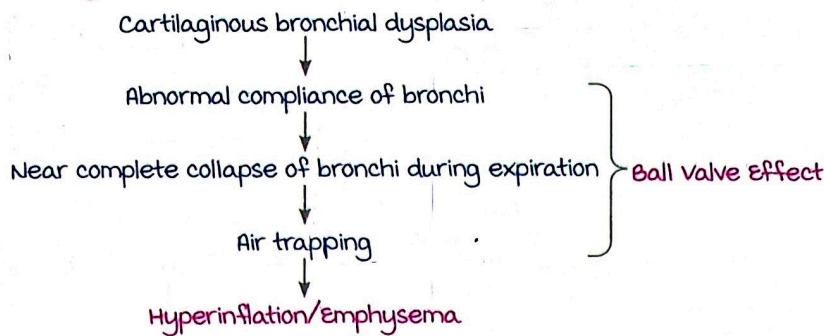
Treatment : **Reassurance.**

- Resolves by 6-12 months of age.

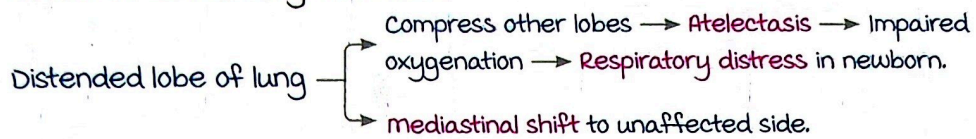


→ **Omega (Ω) shaped epiglottis**

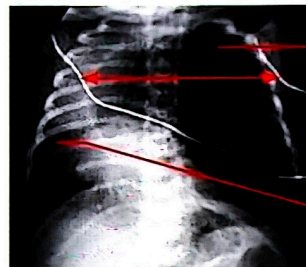
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CONGENITAL LOBAR EMPHYSEMA**Pathogenesis :**

- m/c sites : **Left upper lobe** > Right middle lobe.

Effect on surrounding structures :**Investigation :**

1. Chest Xray :
2. CT chest : **IOC.**



Distension of lung fields

mediastinal shift

Congenital lobar emphysema

Treatment :

- Lobectomy of emphysematous lobe.

CONGENITAL PULMONARY ADENOMATOID MALFORMATION (CPAM)

- Hamartomatous/Dysplastic lung tissue → **multiple cysts (+).**
- m/c cause of **cystic lung disease** in newborn.

Presentation :

Cysts (Non-functional).

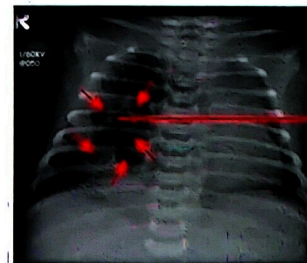
- No gaseous exchange → Respiratory distress.
- ↑ Risk of infection → **Recurrent lung infections.**

Investigation :

1. Chest Xray
2. CT chest : **IOC.**

Treatment :

- ↑ risk of lung tumors → Surgical resection **within one year of age.** (Sarcoma/carcinoma)



multiple cysts

PULMONARY SEQUESTRATION**Pathology :**

- Segregation/isolation of non-functioning part of lung (No communication with bronchus).
 - Blood supply : Systemic (Lower thoracic aorta/abdominal aorta).
↑ risk of recurrent infections.
- Site : Lower lobe of lung fields.

Types :

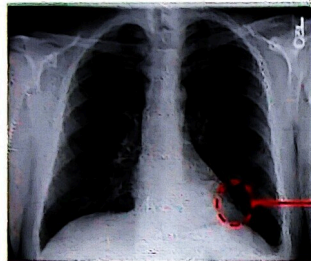
1. Intrapulmonary (Common pleural covering).
2. Extrapulmonary (Separate pleural covering).

Investigation :

1. Chest Xray
2. CT chest : IOC.

Treatment :

Surgical resection.



Sequestered part of lung tissue

Foreign Bodies in the Airway648c85cfee3b03a74e182fab
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Paediatric emergency.

Age group : usually < 3 years.

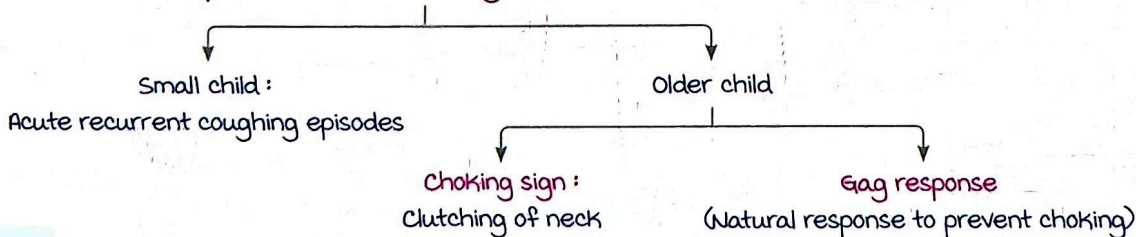
Location : Lower airway > upper airway

m/c : Bronchus (R>L)

Common foreign bodies : Nuts, coins, small rounded plastic toys.

Presentation :

1. Immediate presentation : Choking



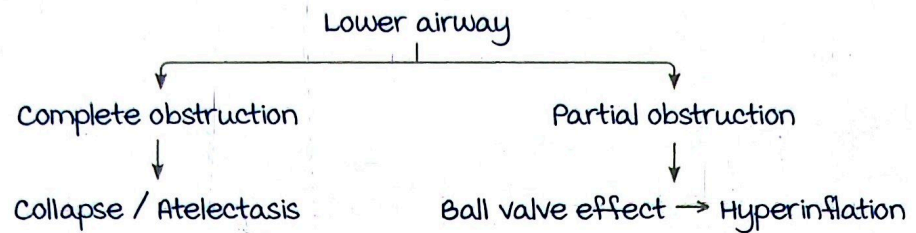
Note : Caregiver may present child with h/o choking.

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2. Delayed presentation : D/t retained foreign body.

- Recurrent unexplained wheeze/breathing difficulty (mimics asthma unresponsive to bronchodilators).
- Suppurative complication.
 - Recurrent pneumonia localising to the same site → Bronchiectasis.
 - Lung abscess.

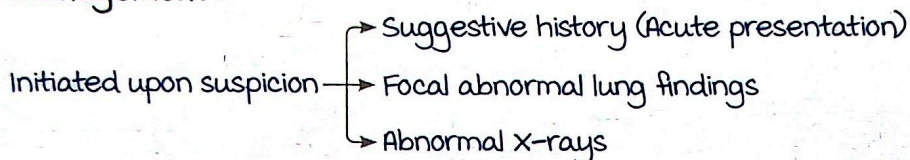
Consequences :



Investigation :

Chest X-ray : Unreliable (Only 10-20% foreign bodies are radiopaque).

management :



1. Lower airway foreign body (Bronchus) → Rigid bronchoscope removal.

2. upper airway foreign body

- > 1 year : Heimlich's maneuver.
- < 1 year : 5 back blows, 5 chest compressions

